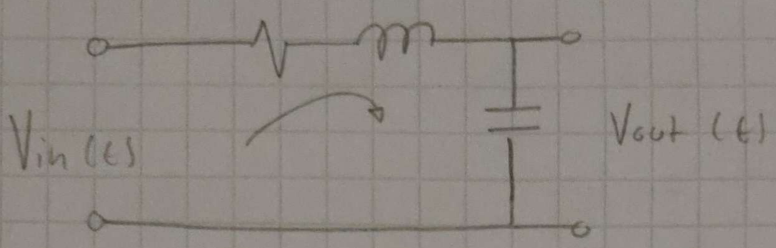


$$V_{in}(t) = i(t)R + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)\,dt$$

$$V_{in}(s) = R \cdot S\,I(s) + L \cdot S^2 I(s) + \frac{1}{C} I(s)$$

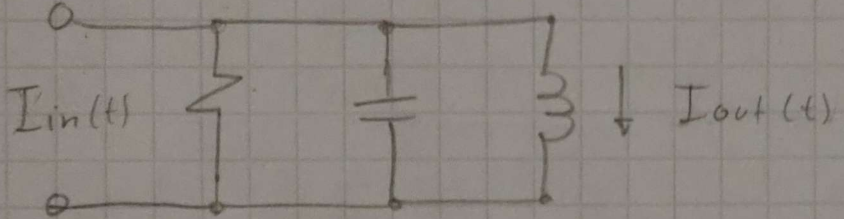
$$V_{in}(s) = \left(LS^2 + \frac{1}{C} + RS\right) \cdot I(s)$$

$$\frac{I(s)}{V_{in}(s)} = \frac{1}{LS^2 + RS + \frac{1}{C}}$$

$$V_c(s) = \frac{1}{C} I(s)$$

$$\boxed{I(s) = C\,V_c(s)} \quad ; \quad V_c = V_{out}$$

b/ $$\frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{CLs^2 + CRs + 1}$$



$$I_{in}(t) = \frac{V(t)}{R} + C\frac{dv(t)}{dt} + I_{out}(t)$$

$$V_L(t) = L\frac{di_L(t)}{dt} \quad ; \quad i_L = I_{out}.$$

Ya que el voltaje es el mismo, reemplazamos:

$$I_{in}(t) = \frac{1}{R}\left(L\frac{di_L(t)}{dt}\right) + C\frac{d}{dt}\left(\frac{di_L(t)}{dt}\right) + I_{out}$$

$$I_{in}(s) = \frac{L}{R} \cdot s\, I_L(s) + LCs^2 I_L(s) + I_L(s)$$

Organizando nos queda:

R/

$$\frac{I_L(s)}{I_{in}(s)} = \frac{1}{LCs^2 + \frac{L}{R}s + 1}$$